source ..... making sound machines
rev 2026.10

stolperbeats

_travelers guide
_for stumbling across the universe

6	24	7
tracks + subdiv	patterns per bank	shuffles × 4 × 4

› then noon = 12 o'clock tags = the panel drawing

shuffled drum trigger sequencer · six tracks + subdiv clock · 7 shuffles × 4 timings × 4 feelings · firmware 162 manual

01 what it is

sb_01

a drum sequencer	trigger patterns for `kick` `snare` `hihat 1` `hihat 2` `perc 1` `perc 2` + a `subdiv` clock
the groove	7 shuffles × 4 timings × 4 feelings, laid over the same patterns
the patterns	24 per bank per track · 64 banks, 4 active out of the box

02 press play

sb_02 · tempo encoder

`[ tempo ]` click	start / stop
`[ tempo ]` turn	15 – 250 bpm in steps of one · nudge to show the tempo
`[ tap ]`	tap quarter notes to set the tempo
`sync` in	a clock here wins over the encoder · the module latches onto its ticks
`reset`	a trigger sends the sequence back to step 1

10 first flight

sb_10 · first beat

1. `kick` `snare` `hihat 1` trigger outs into three drums
2. click `[ tempo ]` to start · turn it to taste
3. turn `[ kick ]` and `[ snare ]` through their 24 patterns
4. turn `[ shuffle ]` down the matrix
5. press `[ shake ]`, then press again for each feeling
6. try `[ trip ]`, `[ push ]` and `[ clave ]` on the same beat

03 patterns

sb_03 · instrument encoders · fig.02

turn	pick one of the 24 patterns in the bank for that track
click	mute / unmute the track
kick · snare	4 lines × 16 steps = 4 bars each
hihats · percs	1 line × 16 steps = 1 bar each, one line per track
one step	a 16th note in 4/4, before shuffle, timing and feeling move it

15 midi in

sb_15 · fixed channels

ch 1 – 6	kick, snare, hihat 1, hihat 2, perc 1, perc 2 · any note steps through the 24 patterns
ch 7	subdiv: shuffled 16th ↔ n-tuplet
ch 8	steps through the 7 shuffles
ch 9	steps through the feelings · every 4 notes moves on a timing

__fig.02 : the pattern matrix

right matrix · 64 / 64 / 4 × 16



16 steps a line · bright dot = the beat · 1 step = a 16th before the shuffle moves it

04 shuffle

sb_04 · shuffle encoder · fig.01

turn	7 shuffles · the further down the left matrix, the stronger
the display	how 8 steps (2 × 4) spread in time · the dots per line are subdivisions
the steps	always 8 per line · only the spacing changes
`shuffle` cv	pick the shuffle with cv

__fig.01 : the shuffle display

left matrix · 7 lines



dots = subdivisions per 4 steps (trip timing) · lower line = stronger shuffle

05 timing · feeling

sb_05 · trip · shake · push · clave

`[ trip ]`	n-tuplet grooves tripping over themselves
`[ shake ]`	golden-age hip hop 8th-note swing
`[ push ]`	gently makes time for the 4/4 downbeats
`[ clave ]`	3:3:2 stays fixed, the rest swings around it
press again	next feeling, shown by the backlight colour: `tight` everything shuffles together
rolls	hihat 2 becomes a trigger burst
flam	kick and snare land very slightly late
slop	flam, with the hihats following

06 edit a pattern

sb_06 · instrument encoders

enter	long press + release that track's encoder
turn · click	move the cursor · set or clear the step
leave	long press the encoder again, then let go …
› fast blink	… now: changes thrown away
› slow blink	… wait for it: changes saved

07 outputs

sb_07 · top row · 10 V

`kick` … `perc 2`	trigger out per track · into a drum's trigger in
`subdiv`	turn the encoder: a shuffled 16th clock or a regular n-tuplet of the shuffle · clock a 1-tick-per-step sequencer
`sync`	unswung clock · ppqn 1 – 10, 12, 16, 24 or once per 1, 2, 4 bars (resets, 4/4)
`env` `env`	two tempo-synced cv outs · section 11

08 clocking

sb_08 · sync encoder

`[ sync ]` click	pick what to edit: in (top row, red) or out (middle row, orange)
sync in	1, 2, 3, 4, 6, 8, 12, 16 or 24 ppqn
first / last pulse	an incoming clock starts the sequence, and stops it when it stops (config)
midi clock	24 ppqn in and out · play sends start / stop

11 envelopes

sb_11 · press + release subdiv 2 s

`env 1`	`[ kick ]` wave · `[ snare ]` clock div · `[ hihat 1 ]` low · `[ hihat 2 ]` high
`env 2`	`[ perc 1 ]` wave · `[ perc 2 ]` clock div · `[ subdiv ]` low · `[ sync ]` high
9 waves	ramp down · ramp up · tri up · tri down · sine · cosine · pulse high · pulse low · stepped
clock div	x4 · x2 · x1 bars · .5 bar · 1 beat · or bd sd h1 h2 p1 p2 = one cycle per hit
:bd … :p2	per-hit velocity · high = accent level, low = non-accent level
leave	press + release subdiv 2 s · saved when you stop the sequence

12 banks

sb_12 · ◄ bank ►

`[ ◄ ]` `[ ► ]`	step through the active banks · 64 banks of 24 patterns
bank edit	press + release a bank button 2 s · same again to leave
`[ kick ]` turn · click	pick a bank (blinking) · set it active or inactive
inactive	skipped by the bank buttons · keeps its patterns
why	perform on a different set · A / B / fill variations · blank slates
`bank` cv	pick the active bank with cv

13 cv · linear

sb_13 · second row

`pattern cv`	one per track · picks a pattern from the active bank
`shuffle` `bank`	cv picks the shuffle and the bank · all 0 – 5 V (jumpers: 10 V)
`[ linear ]`	thins out overlapping steps so one hit plays at a time
who wins	snare › kick › hihat 2 › hihat 1 · snare › kick › perc 2 › perc 1

14 midi out

sb_14 · trs · per-track notes

`midi out`	trs type a · rear switch for b · in takes a or b
7 tracks	channel, note, velocity, duration each
midi edit	hold `[ shuffle ]` 2 s, let go · slow blink
midi rec	hold 3 s · fast blink · incoming midi sets note + channel
show · set	nudge a track encoder · click picks, turn changes
`[ ◄ ]` `[ ► ]`	32 midi banks (drum machine maps) · hold `[ ◄ ]` = save

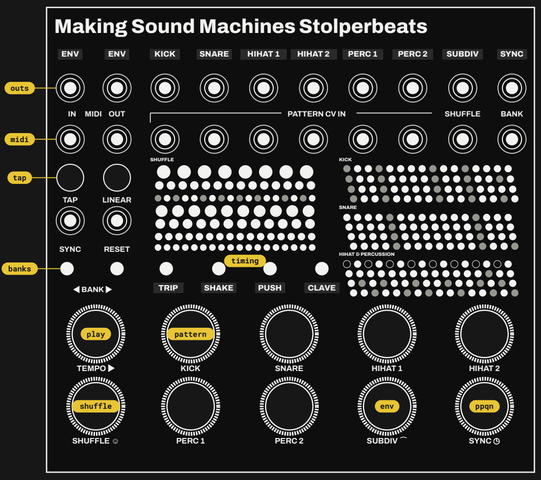
16 card · expander

sb_16

sd card	patterns, settings and midi banks · written when you stop · no card = defaults every power-up
MachineConfig	play-position chaser · clock starts / stops · reset toggles play · midi in, clock and transport on / off
expander	3 hp, passive · linear + mute per track and subdiv · try euclidean or random gates

__fig.03 : the panel

numbers match the legend



`play` `tempo`	click = start / stop · turn = bpm
`pattern` `kick` … `perc 2`	turn = pattern · click = mute · hold = edit
`shuffle` `shuffle`	7 shuffles · hold 2 / 3 s = midi edit / rec
`env` `subdiv`	shuffled 16ths or n-tuplet · hold 2 s = envs
`ppqn` `sync`	clock divider · click = in / out
`timing` `trip` … `clave`	timing · press again = next feeling
`banks` ◄ `bank` ►	active banks · hold 2 s = bank edit
`tap` `tap` · `linear`	tap tempo · one hit at a time
`outs` `top row`	env · triggers · subdiv · sync out
`midi` `second row`	midi in / out · pattern, shuffle, bank cv

09 specs

sb_09

size	26 hp + 3 hp passive expander · 35 mm deep
power	280 mA +12 V · 13 mA –12 V · expander: no power
outs	triggers, clocks and envs at 10 V
cv in	0 – 5 V · rear jumpers for 0 – 10 V
storage	class 10 fat32 sd card · firmware over usb

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel drawing after the making sound machines panel art
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